COMSATS University Islamabad

Department of Statistics

Assignment1

Course Title:	Statistics and Probability Theory			Course Code:	MTH-262	Credit Hours:	3(3,0)
Course Instructor/s:	Miss. Nosheen Amjad			Programme Name:	BCS & BSE		
Semester:		Batch:	BCS & BSE	Section:	A	Date:	
Time Allowed:				Maximum Marks:		10	
Student's Name:				Reg. No.	CUI/		
Important Instructions / Guidelines: Attempt all questions.							

Objective

The objective of this assignment is to help students understand how descriptive statistics can be used to analyze software-related data using SPSS. Students will learn how to compute measures such as mean, median, mode, standard deviation, minimum, maximum, and frequency distribution.

Dataset: Weekly Coding Practice Hours

Student	Coding Hours
1	10
2	12
3	8
4	15
5	18
6	9
7	20
8	14
9	16
10	11
11	13
12	17
13	7
14	19
15	21
16	6
17	12
18	15
19	10
20	14

Part 1: Data Entry in SPSS (4 Marks)

1. Open SPSS software.

2. Go to Variable View and create the following variables:

Variable Name	Type	Label
Student_ID	Numeric	Student Number
Coding_Hours	Numeric	Weekly Coding Practice Hours

3. Switch to Data View and enter the dataset.

Part 2: Descriptive Statistics (8 Marks)

Using Analyze → Descriptive Statistics → Descriptives, calculate the following measures:

- Mean

- Standard Deviation

- Minimum

- Maximum

Questions:

1. What is the average coding practice time of students?

2. What is the highest and lowest number of coding hours?

Part 3: Frequency Distribution (4 Marks)

Using Analyze → Descriptive Statistics → Frequencies:

1. Produce a frequency table.

2. Identify the most common coding hours.

Part 4: Graphical Representation (4 Marks)

Using SPSS charts:

1. Create a Histogram for Coding Hours.

2. Create a Box Plot.

Questions:

1. Is the distribution symmetric or skewed?

2. Are there any outliers?

Submission Requirements

Students must submit:

1. SPSS output screenshots
2. Interpretation (4–5 sentences) explaining the results
3. Graphs generated in SPSS